

DATA VORTEX 2026

ROUND 1 - PHASE 2

SQL ANALYTICS & INSIGHTS REPORT

Selected: 1 Easy Question | 1 Medium Question | 1 Hard Question

Prepared from the project SQLite database using read-only SELECT analysis.

Executive Summary

This submission analyzes 10 leading posts, 33 locations, and 99 location-ranked creator records from the Data Vortex SQLite database. The selected questions progress from post-level engagement (E2), to location aggregation (M1), to within-location creator ranking (H2).

The highest-engagement eligible post is ycjj5zzt7mvx with 7,893 total interactions. The leading location is Los Angeles, USA with 1,603,928 interactions across 400 posts. The highest total among the location leaders belongs to user_zqv2vrf5 in Vancouver, Canada with 81,326 interactions.

EASY QUESTION - E2 - Most Engaged Posts

Official question: Find the top 10 posts based on total engagement. Total engagement is defined as likes + shares + comments. Ignore posts where likes are missing.

Method

Post-level totals expose the individual content items that generated the strongest complete engagement.

Executed Output

post_id	user_id	platform	likes	shares	comments	total_engagement
ycjj5zzt7mvx	user_d9971ba6	Instagram	4,983	1,919	991	7,893
wo7py9aljg3t	user_o8le7hqf	Reddit	4,864	1,981	948	7,793
gmoeib832zbs	user_pe5yckyby	Facebook	4,902	1,880	982	7,764
5kvuyvf38nqx	user_z0feut2e	YouTube	4,923	1,971	861	7,755
pvfl3d8hj7jd	user_csluibwk	Instagram	4,989	1,840	909	7,738
tdgjyylpua20	user_8nvzxsuj		4,979	1,932	812	7,723
tne7s3o4l4wd	user_lr3fagdl	Instagram	4,931	1,903	878	7,712
a1kiwl618kzy	user_aaiari8o	Facebook	4,811	1,952	920	7,683
fp89q1ickn9w	user_h4lueh1i	Twitter	4,740	1,933	955	7,628
5n161ir5hhhr	user_u98jwp3f	YouTube	4,751	1,981	878	7,610

Interpretation

All displayed values come directly from the executed SELECT query. Posts with missing likes are excluded so that every engagement total uses all three requested components.

MEDIUM QUESTION - M1 - Which Locations Generate the Most Engagement?

Official question: Using both datasets, calculate the total engagement generated by users from each location. Return the location, number of posts, and total engagement. Rank locations from highest to lowest engagement.

Method

Location aggregation compares creator communities while preserving each post exactly once through the users-to-posts relationship.

Executed Output

location	post_count	total_engagement
Los Angeles, USA	400	1,603,928
Munich, Germany	389	1,555,610
Barcelona, Spain	372	1,528,127
Shanghai, China	382	1,522,852
Dubai, UAE	351	1,428,959
Rio de Janeiro, Brazil	360	1,415,136
Melbourne, Australia	346	1,412,196
Houston, USA	355	1,400,008
Mumbai, India	354	1,376,637
Osaka, Japan	331	1,372,085
London, UK	334	1,352,812
Milan, Italy	339	1,345,355
New York, USA	325	1,325,029
Johannesburg, South Africa	337	1,322,131
Chicago, USA	322	1,297,999
Beijing, China	315	1,281,335
São Paulo, Brazil	325	1,274,137
Paris, France	314	1,272,108
Berlin, Germany	308	1,242,015
Singapore	297	1,200,424
Tokyo, Japan	293	1,187,546
Toronto, Canada	290	1,164,704
Manchester, UK	283	1,128,627
Vancouver, Canada	287	1,117,980
Rome, Italy	284	1,102,222
Mexico City, Mexico	271	1,101,899
Cairo, Egypt	264	1,077,877
Delhi, India	254	1,024,042
Madrid, Spain	246	983,352
Seoul, South Korea	243	944,756
Lyon, France	248	934,458
Lagos, Nigeria	193	753,250
Sydney, Australia	174	714,642

Interpretation

All displayed values come directly from the executed SELECT query. Posts with missing likes are excluded so that every engagement total uses all three requested components.

HARD QUESTION - H2 - Rank Users Within Their Location

Official question: For every location, rank users based on their total engagement. Return only the top 3 users from each location.

Method

Partitioned ranking distinguishes local leadership from a single global leaderboard and therefore identifies leading creators in every represented location.

Executed Output

location	engagement_rank	user_id	follower_count	post_count	total_engagement
Barcelona, Spain	1	user_nfo3ih5u	40,429	17	69,109
Barcelona, Spain	2	user_5s9ifp0y	28,412	10	55,663
Barcelona, Spain	3	user_24wzfb8b	38,633	12	49,741
Beijing, China	1	user_cdzp4vm2	11,187	11	48,019
Beijing, China	2	user_ttlouvlq	19,373	11	45,695
Beijing, China	3	user_gre0h06x	23,303	10	43,719
Berlin, Germany	1	user_6e99lerx	25,366	13	52,982
Berlin, Germany	2	user_rtp2dykx	14,113	11	52,748
Berlin, Germany	3	user_m5z8a3sg	34,800	11	50,121
Cairo, Egypt	1	user_g37hrbp3	6,699	10	50,320
Cairo, Egypt	2	user_cuzig14g	8,070	12	48,076
Cairo, Egypt	3	user_ogtvuuki	4,459	8	41,183
Chicago, USA	1	user_ujljl1n7	49,855	13	59,050
Chicago, USA	2	user_pr7bcf45	30,128	13	57,393
Chicago, USA	3	user_u98jwp3f	49,936	10	46,083
Delhi, India	1	user_zq2nib62	5,153	12	51,134
Delhi, India	2	user_8776jund	21,147	10	48,425
Delhi, India	3	user_yu2w96i5	9,772	13	45,589
Dubai, UAE	1	user_xetn4exw	14,902	15	57,883
Dubai, UAE	2	user_wyeb78qv	24,634	16	55,214
Dubai, UAE	3	user_rsrq2sgy	32,791	11	54,042
Houston, USA	1	user_r7eg1rac	2,052	12	54,552
Houston, USA	2	user_68enpikx	9,364	11	50,580
Houston, USA	3	user_44h4xl31	48,691	10	45,515
Johannesburg, South Africa	1	user_9mtets0p	8,262	14	62,414
Johannesburg, South Africa	2	user_6wra58f7	4,953	13	53,587
Johannesburg, South Africa	3	user_cefi5hpb	15,895	10	48,454
Lagos, Nigeria	1	user_slqlepta	40,296	13	57,975
Lagos, Nigeria	2	user_jnknxe13	35,496	12	43,962
Lagos, Nigeria	3	user_sk29dqyu	18,327	11	42,902
London, UK	1	user_634lriof	12,327	12	60,436
London, UK	2	user_94hehp26	33,258	13	59,706
London, UK	3	user_3jxx3h3w	35,570	12	50,074
Los Angeles, USA	1	user_3zzvv8vp	42,908	13	59,221
Los Angeles, USA	2	user_8wk0j5ae	36,943	16	58,745
Los Angeles, USA	3	user_wcqtxwgr	10,281	11	51,432

location	engagement_rank	user_id	follower_count	post_count	total_engagement
Lyon, France	1	user_fgjkkrie	2,211	14	62,690
Lyon, France	2	user_2lslaaod	7,105	7	36,315
Lyon, France	3	user_ta69gmj0	21,267	11	35,427
Madrid, Spain	1	user_v4objqex	16,772	11	47,252
Madrid, Spain	2	user_edi9lmxp	22,782	12	46,818
Madrid, Spain	3	user_w4p8zi5g	7,103	11	44,392
Manchester, UK	1	user_oknxkzax	34,257	13	57,292
Manchester, UK	2	user_q3vs0uj7	32,714	14	51,798
Manchester, UK	3	user_ouggehu2	29,402	10	43,732
Melbourne, Australia	1	user_d6tjszz8	37,186	14	58,020
Melbourne, Australia	2	user_qdztdeac	14,266	11	55,609
Melbourne, Australia	3	user_mtuzxdlb	49,341	12	51,825
Mexico City, Mexico	1	user_716l1rz5	41,236	12	54,592
Mexico City, Mexico	2	user_s87enj6b	43,476	11	53,403
Mexico City, Mexico	3	user_cyqqg8x5	44,679	11	47,637
Milan, Italy	1	user_xa1s1sd5	37,758	12	51,002
Milan, Italy	2	user_rr1uzkql	1,069	12	46,912
Milan, Italy	3	user_t59clgrj	43,441	10	43,406
Mumbai, India	1	user_2ytzut7i	17,064	13	63,888
Mumbai, India	2	user_y9bjho1	24,858	13	61,956
Mumbai, India	3	user_gq0rypar	17,672	13	60,450
Munich, Germany	1	user_al9p1gnu	49,268	11	52,575
Munich, Germany	2	user_g9g7onlw	7,981	11	48,735
Munich, Germany	3	user_58rc29a2	29,379	11	44,526
New York, USA	1	user_wr2785sn	12,713	11	53,050
New York, USA	2	user_kk1qdsq3	32,189	11	47,175
New York, USA	3	user_tlwy34pg	24,453	9	46,737
Osaka, Japan	1	user_yr80oaxg	30,039	11	49,062
Osaka, Japan	2	user_c66o14p3	24,996	10	47,727
Osaka, Japan	3	user_5r58l1kr	23,520	9	45,641
Paris, France	1	user_kf84zww2	49,575	12	51,150
Paris, France	2	user_45yefmlp	37,589	10	47,846
Paris, France	3	user_4snqim59	20,152	11	41,990
Rio de Janeiro, Brazil	1	user_hdas0iau	1,620	14	55,118
Rio de Janeiro, Brazil	2	user_aroc3ncd	10,428	12	54,206
Rio de Janeiro, Brazil	3	user_31qdm3s9	33,635	11	52,060
Rome, Italy	1	user_uerv85na	1,824	13	58,861
Rome, Italy	2	user_0aacyvtz	1,452	10	44,667
Rome, Italy	3	user_txrypced	2,479	10	42,896
Seoul, South Korea	1	user_ceim4vux	40,811	12	45,157
Seoul, South Korea	2	user_tmjtxubu	48,104	10	44,602
Seoul, South Korea	3	user_ivpoy0x1	28,950	11	43,343
Shanghai, China	1	user_zyptgxi8	47,455	11	51,627
Shanghai, China	2	user_xn2a0g8z	45,029	10	44,370
Shanghai, China	3	user_qq52pl85	23,934	10	44,009

location	engagement_rank	user_id	follower_count	post_count	total_engagement
Singapore	1	user_14ibkds2	7,361	13	57,044
Singapore	2	user_v469etxv	23,469	11	44,636
Singapore	3	user_nudota2x	17,212	11	44,092
Sydney, Australia	1	user_bnswpe65	20,002	14	54,984
Sydney, Australia	2	user_h4lueh1i	17,902	9	43,948
Sydney, Australia	3	user_kiebkz7i	34,969	9	39,205
São Paulo, Brazil	1	user_0snf4coc	42,927	14	54,457
São Paulo, Brazil	2	user_8bcte4x2	46,936	12	51,993
São Paulo, Brazil	3	user_e9paascb	19,695	14	48,463
Tokyo, Japan	1	user_d4eat3v3	49,914	14	62,588
Tokyo, Japan	2	user_kbdvf8d6	4,645	10	48,490
Tokyo, Japan	3	user_csluibwk	19,091	11	48,358
Toronto, Canada	1	user_0irp4abu	44,206	14	64,133
Toronto, Canada	2	user_cpxks4r	49,247	12	49,658
Toronto, Canada	3	user_ukkkyp0o	28,045	9	42,441
Vancouver, Canada	1	user_zqv2vrf5	13,531	19	81,326
Vancouver, Canada	2	user_j1e1kek5	46,765	15	61,112
Vancouver, Canada	3	user_lk8wtuqr	32,276	14	51,577

Interpretation

All displayed values come directly from the executed SELECT query. Posts with missing likes are excluded so that every engagement total uses all three requested components.

Cross-Question Insights

The analyses operate at complementary levels: individual posts identify exceptional content, location totals quantify collective output, and the within-location ranking exposes the creators responsible for local leadership. Because all three use the same complete-engagement rule, their totals are directly comparable without silently treating missing likes as zero.

Data Quality and Validation

The source database contains 1,500 users and 12,000 posts. SQLite integrity and foreign-key checks pass. The analysis used a read-only connection; no INSERT, UPDATE, DELETE, DROP, or ALTER statement was issued. Location totals were reconciled to a direct complete-engagement total, and the location-rank output was checked for valid top-three ranks.

Limitations

Likes are missing for a subset of posts. Rather than impute a value, this submission excludes those posts whenever an exact total of likes, shares, and comments is required. The findings describe this dataset and do not establish causal effects of platform, location, or follower count.

Conclusion

The selected SQL solutions provide a validated view from high-performing posts through location-level contribution to creator leadership. Together, they demonstrate reproducible aggregation, relational joins, and window-function ranking on the production Data Vortex database.